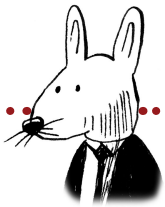
learning multiple weights at a time: generalizing gradient descent

5



In this chapter

- Gradient descent learning with multiple inputs
- Freezing one weight: what does it do?
- Gradient descent learning with multiple outputs
- Gradient descent learning with multiple inputs and outputs
- Visualizing weight values
- Visualizing dot products

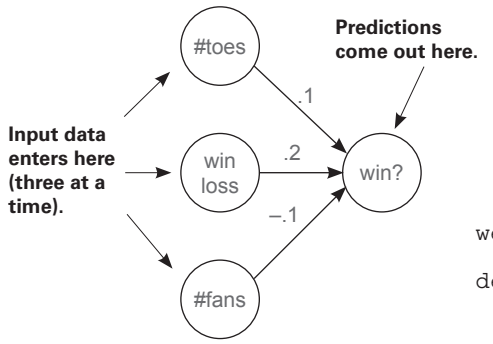
> *You don't learn to walk by following rules. You learn by doing and by falling over.*
>
> —Richard Branson, http://mng.bz/oYgd

Gradient descent learning with multiple inputs

Gradient descent also works with multiple inputs.

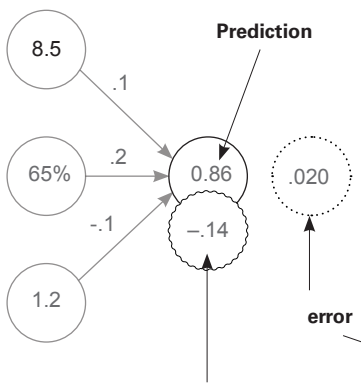
In the preceding chapter, you learned how to use gradient descent to update a weight. In this chapter, we'll more or less reveal how the same techniques can be used to update a network that contains multiple weights. Let's start by jumping in the deep end, shall we? The following diagram shows how a network with multiple inputs can learn.

❶ An empty network with multiple inputs




```
def w_sum(a,b):
    assert(len(a) == len(b))
    output = 0
    for i in range(len(a)):
        output += (a[i] * b[i])
    return output

weights = [0.1, 0.2, -.1]

def neural_network(input, weights):
    pred = w_sum(input,weights)
    return pred
```

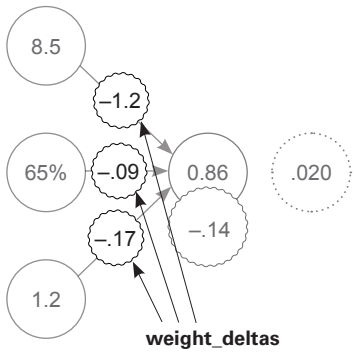
❷ PREDICT + COMPARE: Making a prediction, and calculating error and delta



Input corresponds to every entry for the first game of the season.

```
toes =  [8.5 , 9.5, 9.9, 9.0]
wlrec = [0.65, 0.8, 0.8, 0.9]
nfans = [1.2 , 1.3, 0.5, 1.0]

win_or_lose_binary = [1, 1, 0, 1]

true = win_or_lose_binary[0]

input = [toes[0],wlrec[0],nfans[0]]

pred = neural_network(input,weights)

error = (pred - true) ** 2

delta = pred - true
```

❸ LEARN: Calculating each weight_delta and putting it on each weight

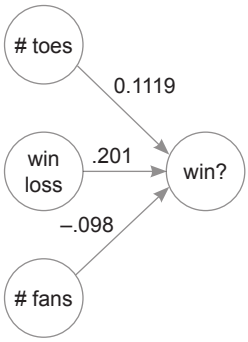



```
def ele_mul(number,vector):

    output = [0,0,0]

    assert(len(output) == len(vector))

    for i in range(len(vector)):
        output[i] = number * vector[i]

    return output

input = [toes[0],wlrec[0],nfans[0]]

pred = neural_network(input,weight)

error = (pred - true) ** 2

delta = pred - true

weight_deltas = ele_mul(delta,input)
```

```
8.5 * -0.14 = -1.19  = weight_deltas[0]
0.65 * -0.14 = -0.091 = weight_deltas[1]
1.2 * -0.14 = -0.168 = weight_deltas[2]
```

There's nothing new in this diagram. Each `weight_delta` is calculated by taking its output `delta` and multiplying it by its `input`. In this case, because the three weights share the same output node, they also share that node's `delta`. But the weights have different weight `deltas` owing to their different `input` values. Notice further that you can reuse the `ele_mul` function from before, because you're multiplying each value in `weights` by the same value `delta`.

❹ LEARN: Updating the weights




```
input = [toes[0],wlrec[0],nfans[0]]

pred = neural_network(input,weight)
error = (pred - true) ** 2
delta = pred - true

weight_deltas = ele_mul(delta,input)

alpha = 0.01

for i in range(len(weights)):
    weights[i] -= alpha * weight_deltas[i]
print("Weights:" + str(weights))
print("Weight Deltas:" + str(weight_deltas))
```

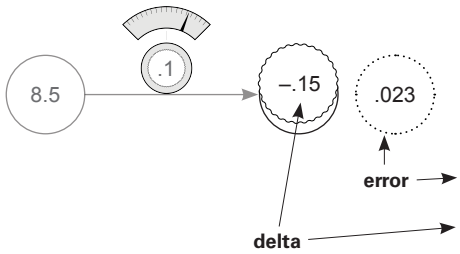
```
0.1 - (-1.19 * 0.01) = 0.1119  = weights[0]
0.2 - (-.091 * 0.01) = 0.2009  = weights[1]
-0.1 - (-.168 * 0.01) = -0.098  = weights[2]
```

Gradient descent with multiple inputs explained

Simple to execute, and fascinating to understand.

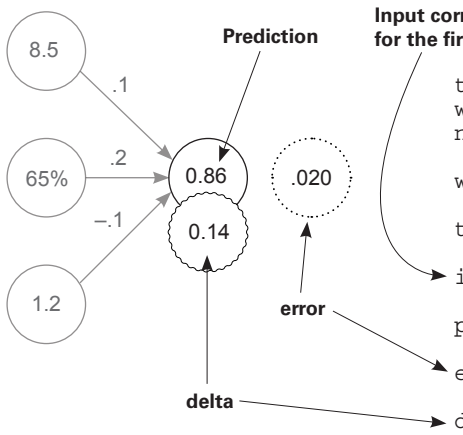
When put side by side with the single-weight neural network, gradient descent with multiple inputs seems rather obvious in practice. But the properties involved are fascinating and worthy of discussion. First, let's take a look at them side by side.

❶ Single input: Making a prediction and calculating error and delta




```
number_of_toes = [8.5]
win_or_lose_binary = [1] # (won!!!)

input = number_of_toes[0]
true = win_or_lose_binary[0]

pred = neural_network(input,weight)

error = (pred - true) ** 2

delta = pred - true
```

❷ Multi-input: Making a prediction and calculating error and delta



Input corresponds to every entry for the first game of the season

```
toes =  [8.5, 9.5, 9.9, 9.0]
wlrec = [0.65, 0.8, 0.8, 0.9]
nfans = [1.2, 1.3, 0.5, 1.0]

win_or_lose_binary = [1, 1, 0, 1]

true = win_or_lose_binary[0]

input = [toes[0],wlrec[0],nfans[0]]

pred = neural_network(input,weights)

error = (pred - true) ** 2

delta = pred - true
```

Up until the generation of `delta` on the output node, single input and multi-input gradient descent are identical (other than the prediction differences we studied in chapter 3). You make a prediction and calculate `error` and `delta` in identical ways. But the following problem remains: when you had only one `weight`, you had only one `input` (one `weight_delta` to generate). Now you have three. How do you generate three `weight_deltas`?

How do you turn a single delta (on the node) into three weight_delta values?

Remember the definition and purpose of `delta` versus `weight_delta`. `delta` is a measure of how much you want a node's value to be different. In this case, you compute it by a direct subtraction between the node's value and what you wanted the node's value to be (`pred - true`). Positive `delta` indicates the node's value was too high, and negative that it was too low.

> **delta**
>
> A measure of how much higher or lower you want a node's value to be, to predict perfectly given the current training example.

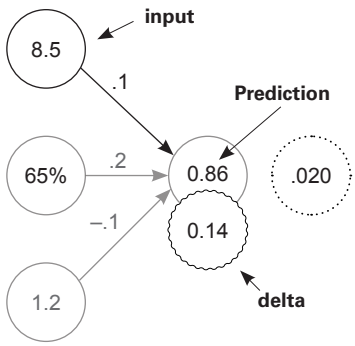
`weight_delta`, on the other hand, is an *estimate* of the direction and amount to move the weights to reduce `node_delta`, inferred by the derivative. How do you transform `delta` into a `weight_delta`? You multiply `delta` by a weight's `input`.

> **weight_delta**
>
> A derivative-based estimate of the direction and amount you should move a weight to reduce `node_delta`, accounting for scaling, negative reversal, and stopping.

Consider this from the perspective of a single weight, highlighted at right:

`delta`: Hey, `inputs`—yeah, you three. Next time, predict a little higher.

Single weight: Hmm: if my `input` was 0, then my weight wouldn't have mattered, and I wouldn't change a thing (*stopping*). If my `input` was negative, then I'd want to decrease my weight instead of increase it (*negative reversal*). But my `input` is positive and quite large, so I'm *guessing* that my personal prediction mattered a lot to the aggregated output. I'm going to move my weight up a lot to compensate (*scaling*).



The single weight increases its value.

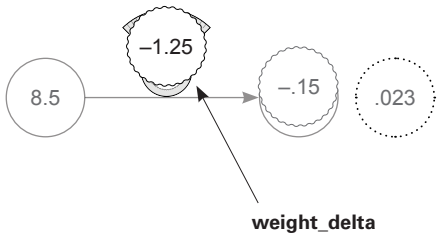
What did those three properties/statements really say? They all (stopping, negative reversal, and scaling) made an observation of how the weight's role in `delta` was affected by its `input`. Thus, each `weight_delta` is a sort of input-modified version of `delta`.

This brings us back to the original question: how do you turn one (node) `delta` into three `weight_delta` values? Well, because each weight has a unique input and a shared `delta`, you

use each respective weight's `input` multiplied by `delta` to create each respective `weight_delta`. Let's see this process in action.

In the next two figures, you can see the generation of `weight_delta` variables for the previous single-input architecture and for the new multi-input architecture. Perhaps the easiest way to see how similar they are is to read the pseudocode at the bottom of each figure. Notice that the multi-weight version multiplies `delta` (0.14) by every input to create the various `weight_delta`s. It's a simple process.

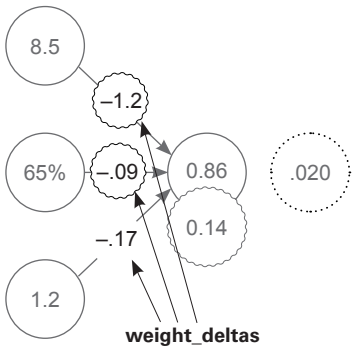
❸ Single input: Calculating weight_delta and putting it on the weight




```
number_of_toes = [8.5]
win_or_lose_binary = [1] # (won!!!)

input = number_of_toes[0]
true = win_or_lose_binary[0]

pred = neural_network(input,weight)

error = (pred - true) ** 2

delta = pred - true

weight_delta = input * delta
```

8.5 * –0.15 = –1.25 => **weight_delta**

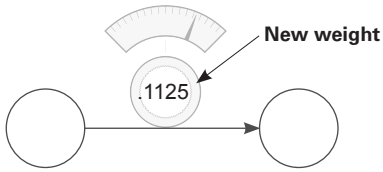
❹ Multi-input: Calculating each weight_delta and putting it on each weight




```
def ele_mul(number,vector):

    output = [0,0,0]

    assert(len(output) == len(vector))

    for i in range(len(vector)):
        output[i] = number * vector[i]

    return output

input = [toes[0],wlrec[0],nfans[0]]

pred = neural_network(input,weights)

error = (pred - true) ** 2

delta = pred - true

weight_deltas = ele_mul(delta,input)
```

8.5 * 0.14 = –1.2 => **weight_deltas[0]**
0.65 * 0.14 = –.09 => **weight_deltas[1]**
1.2 * 0.14 = –.17 => **weight_deltas[2]**

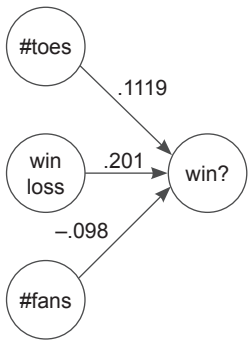
5 Updating the weight



You multiply `weight_delta` by a small number, `alpha`, before using it to update the weight. This allows you to control how quickly the network learns. If it learns too quickly, it can update weights too aggressively and overshoot. Note that the weight update made the same change (small increase) as hot and cold learning.

```
number_of_toes = [8.5]
win_or_lose_binary = [1] # (won!!!)

input = number_of_toes[0]
true = win_or_lose_binary[0]

pred = neural_network(input,weight)

error = (pred - true) ** 2

delta = pred - true

weight_delta = input * delta

alpha = 0.01          ← Fixed before training

weight -= weight_delta * alpha
```

6 Updating the weights




```
input = [toes[0],wlrec[0],nfans[0]]

pred = neural_network(input,weights)

error = (pred - true) ** 2

delta = pred - true

weight_deltas = ele_mul(delta,input)

alpha = 0.01

for i in range(len(weights)):
    weights[i] -= alpha * weight_deltas[i]
```

0.1 – (1.19 * 0.01) = 0.1119 = **weights[0]**
0.2 – (.091 * 0.01) = 0.2009 = **weights[1]**
–0.1 – (.168 * 0.01) = –0.098 = **weights[2]**

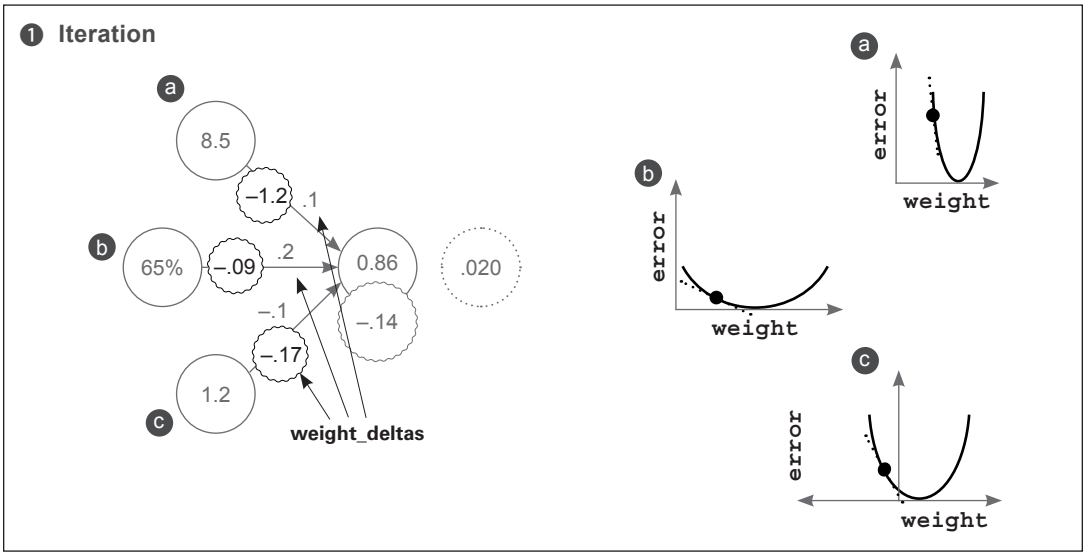
The last step is also nearly identical to the single-input network. Once you have the `weight_delta` values, you multiply them by `alpha` and subtract them from the weights. It's literally the same process as before, repeated across multiple weights instead of a single one.

Let's watch several steps of learning

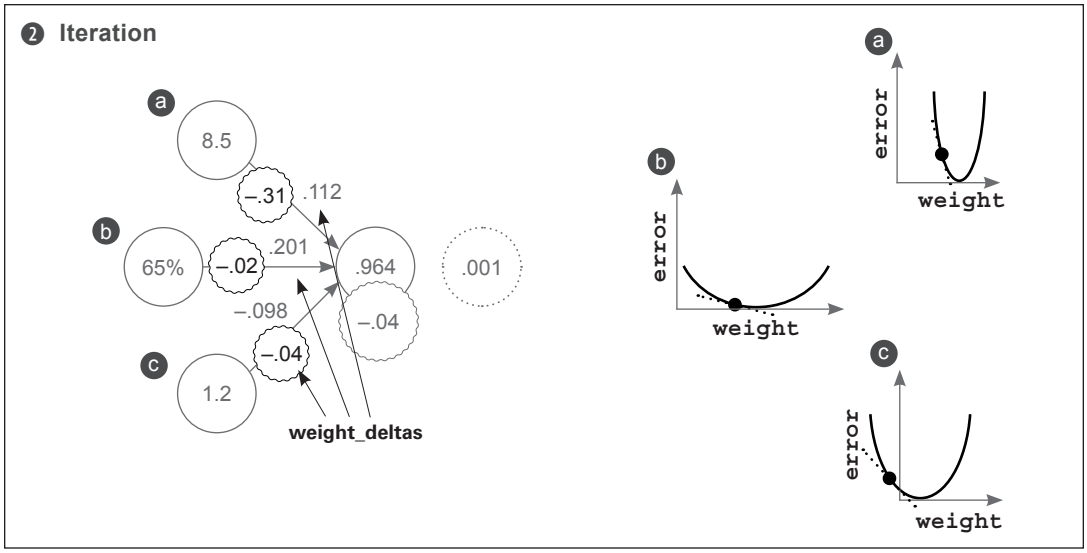
```
def neural_network(input, weights):
  out = 0
  for i in range(len(input)):
    out += (input[i] * weights[i])
  return out

def ele_mul(scalar, vector):
  out = [0,0,0]
  for i in range(len(out)):
    out[i] = vector[i] * scalar
  return out

toes =  [8.5, 9.5, 9.9, 9.0]
wlrec = [0.65, 0.8, 0.8, 0.9]
nfans = [1.2, 1.3, 0.5, 1.0]

win_or_lose_binary = [1, 1, 0, 1]
true = win_or_lose_binary[0]

alpha = 0.01
weights = [0.1, 0.2, -.1]
input = [toes[0],wlrec[0],nfans[0]]
```

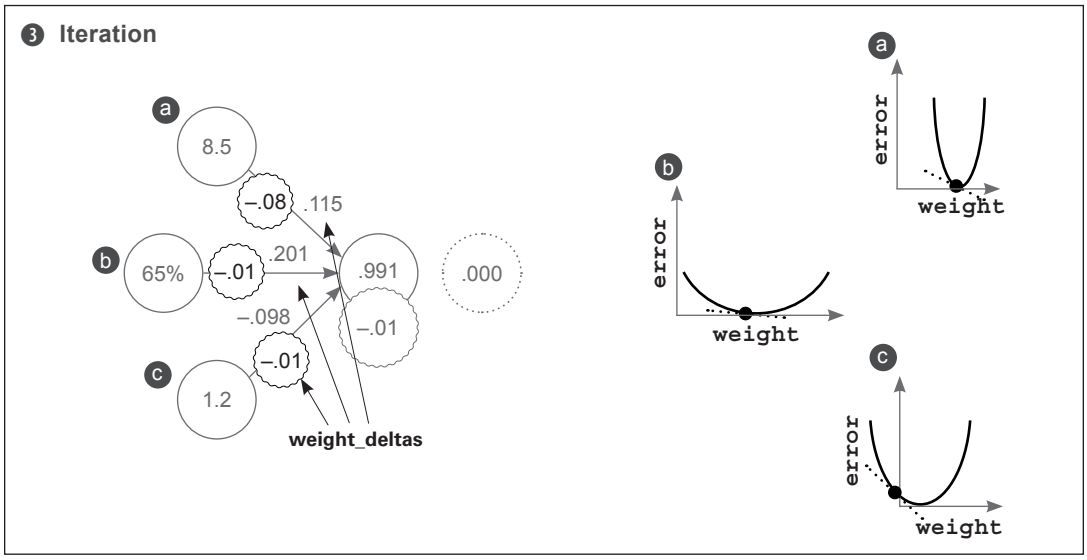
(*continued*)

```
for iter in range(3):

  pred = neural_network(input,weights)

  error = (pred - true) ** 2
  delta = pred - true

  weight_deltas=ele_mul(delta,input)

  print("Iteration:" + str(iter+1))
  print("Pred:" + str(pred))
  print("Error:" + str(error))
  print("Delta:" + str(delta))
  print("Weights:" + str(weights))
  print("Weight_Deltas:")
  print(str(weight_deltas))
  print(
  )

  for i in range(len(weights)):
    weights[i]-=alpha*weight_deltas[i]
```



We can make three individual error/weight curves, one for each weight. As before, the slopes of these curves (the dotted lines) are reflected by the `weight_delta` values. Notice that a is steeper than the others. Why is `weight_delta` steeper for a than the others if they share the same output `delta` and `error` measure? Because a has an `input` value that's significantly higher than the others and thus, a higher derivative.





Here are a few additional takeaways. Most of the learning (weight changing) was performed on the weight with the largest input ⓐ, because the input changes the slope significantly. This isn't necessarily advantageous in all settings. A subfield called *normalization* helps encourage learning across all weights despite dataset characteristics such as this. This significant difference in slope forced me to set `alpha` lower than I wanted (0.01 instead of 0.1). Try setting `alpha` to 0.1: do you see how ⓐ causes it to diverge?

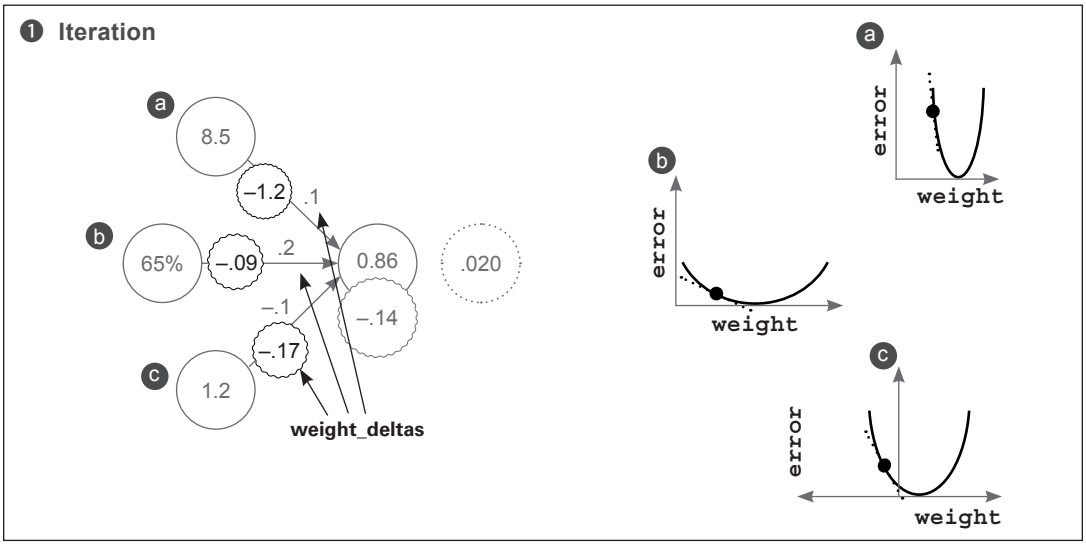
Freezing one weight: What does it do?

This experiment is a bit advanced in terms of theory, but I think it's a great exercise to understand how the weights affect each other. You're going to train again, except weight ⓐ won't ever be adjusted. You'll try to learn the training example using only weights ⓑ and ⓒ (`weights[1]` and `weights[2]`).

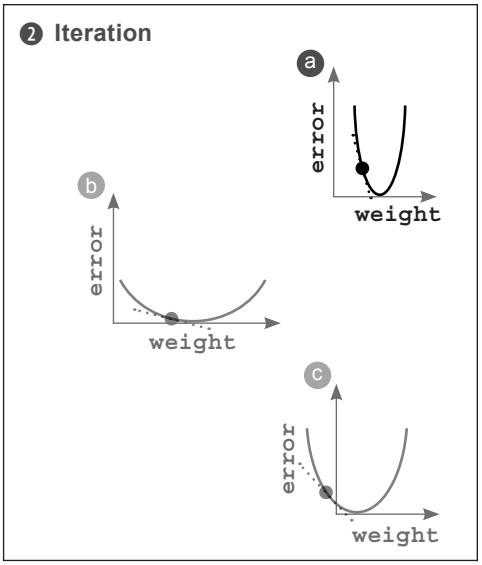
```
def neural_network(input, weights):
  out = 0
  for i in range(len(input)):
    out += (input[i] * weights[i])
  return out

def ele_mul(scalar, vector):
  out = [0,0,0]
  for i in range(len(out)):
    out[i] = vector[i] * scalar
  return out

toes =  [8.5, 9.5, 9.9, 9.0]
wlrec = [0.65, 0.8, 0.8, 0.9]
nfans = [1.2, 1.3, 0.5, 1.0]

win_or_lose_binary = [1, 1, 0, 1]
true = win_or_lose_binary[0]

alpha = 0.3
weights = [0.1, 0.2, -.1]
input = [toes[0],wlrec[0],nfans[0]]
```

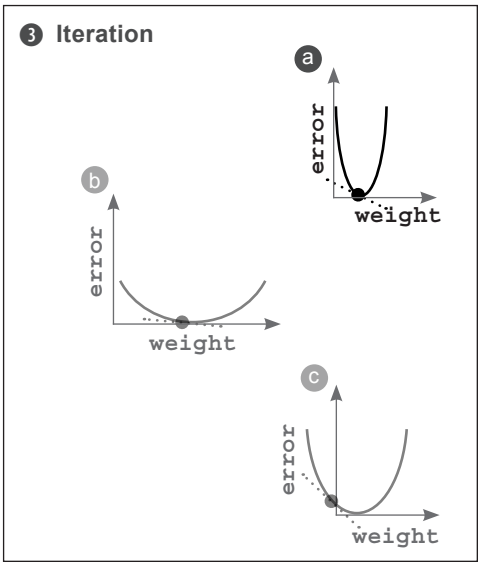
(*continued*)

```
for iter in range(3):

  pred = neural_network(input,weights)

  error = (pred - true) ** 2
  delta = pred - true

  weight_deltas=ele_mul(delta,input)
  weight_deltas[0] = 0

  print("Iteration:" + str(iter+1))
  print("Pred:" + str(pred))
  print("Error:" + str(error))
  print("Delta:" + str(delta))
  print("Weights:" + str(weights))
  print("Weight_Deltas:")
  print(str(weight_deltas))
  print(
  )

  for i in range(len(weights)):
    weights[i]-=alpha*weight_deltas[i]
```



Perhaps you're surprised to see that ⓐ still finds the bottom of the bowl. Why is this? Well, the curves are a measure of each individual weight relative to the global error. Thus, because `error` is shared, when one weight finds the bottom of the bowl, all the weights find the bottom of the bowl.





This is an extremely important lesson. First, if you converged (reached `error` = 0) with ⓑ and ⓒ weights and then tried to train ⓐ, ⓐ wouldn't move. Why? `error` = 0, which means `weight_delta` is 0. This reveals a potentially damaging property of neural networks: ⓐ may be a powerful input with lots of predictive power, but if the network accidentally figures out how to predict accurately on the training data without it, then it will never learn to incorporate ⓐ into its prediction.

Also notice how ⓐ finds the bottom of the bowl. Instead of the black dot moving, the curve seems to move to the left. What does this mean? The black dot can move horizontally only if the weight is updated. Because the weight for ⓐ is frozen for this experiment, the dot must stay fixed. But `error` clearly goes to 0.

This tells you what the graphs really are. In truth, these are 2D slices of a four-dimensional shape. Three of the dimensions are the weight values, and the fourth dimension is the error. This shape is called the *error plane*, and, believe it or not, its curvature is determined by the training data. Why is that the case?

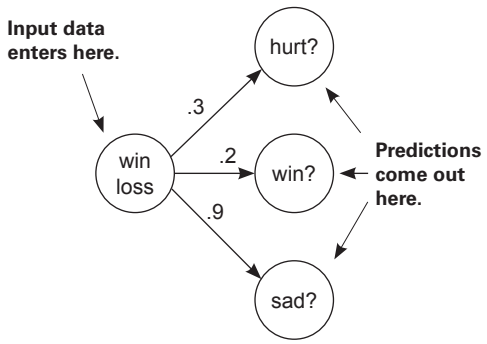
`error` is determined by the training data. Any network can have any `weight` value, but the value of `error` given any particular weight configuration is 100% determined by data. You've already seen how the steepness of the U shape is affected by the input data (on several occasions). What you're really trying to do with the neural network is find the lowest point on this big error plane, where the lowest point refers to the lowest `error`. Interesting, eh? We'll come back to this idea later, so file it away for now.

Gradient descent learning with multiple outputs

Neural networks can also make multiple predictions using only a single input.

Perhaps this will seem a bit obvious. You calculate each `delta` the same way and then multiply them all by the same, single input. This becomes each weight's `weight_delta`. At this point, I hope it's clear that a simple mechanism (stochastic gradient descent) is consistently used to perform learning across a wide variety of architectures.

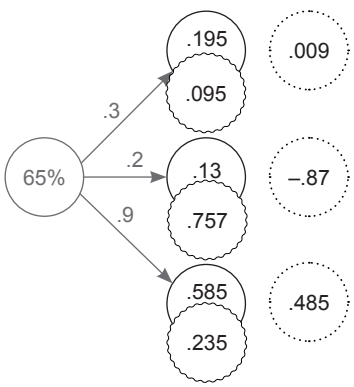
❶ An empty network with multiple outputs



Instead of predicting just whether the team won or lost, now you're also predicting whether they're happy or sad *and* the percentage of the team members who are hurt. You're making this prediction using only the current win/loss record.

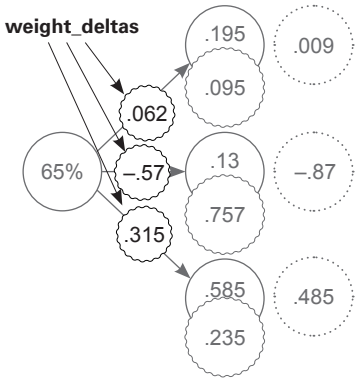
```
weights = [0.3, 0.2, 0.9]

def neural_network(input, weights):

    pred = ele_mul(input,weights)

    return pred
```

❷ PREDICT: Making a prediction and calculating error and delta




```
wlrec = [0.65, 1.0, 1.0, 0.9]

hurt  = [0.1, 0.0, 0.0, 0.1]
win   = [  1,   1,   0,   1]
sad   = [0.1, 0.0, 0.1, 0.2]

input = wlrec[0]
true = [hurt[0], win[0], sad[0]]

pred = neural_network(input,weights)

error = [0, 0, 0]
delta = [0, 0, 0]

for i in range(len(true)):

    error[i] = (pred[i] - true[i]) ** 2
    delta[i] = pred[i] - true[i]
```

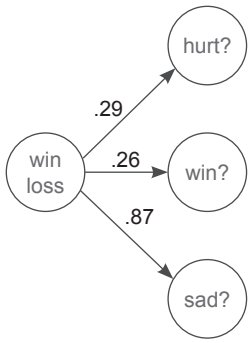
❸ COMPARE: Calculating each weight_delta and putting it on each weight




```
def scalar_ele_mul(number,vector):

    output = [0,0,0]

    assert(len(output) == len(vector))

    for i in range(len(vector)):
        output[i] = number * vector[i]

    return output

wlrec = [0.65, 1.0, 1.0, 0.9]

hurt  = [0.1, 0.0, 0.0, 0.1]
win   = [  1,   1,   0,   1]
sad   = [0.1, 0.0, 0.1, 0.2]

input = wlrec[0]
true = [hurt[0], win[0], sad[0]]

pred = neural_network(input,weights)

error = [0, 0, 0]
delta = [0, 0, 0]

for i in range(len(true)):

    error[i] = (pred[i] - true[i]) ** 2
    delta[i] = pred[i] - true[i]

weight_deltas = scalar_ele_mul(input,weights)
```

As before, `weight_deltas` are computed by multiplying the input node value with the output node `delta` for each weight. In this case, the `weight_deltas` share the same input node and have unique output nodes (`deltas`). Note also that you can reuse the `ele_mul` function.

❹ LEARN: Updating the weights

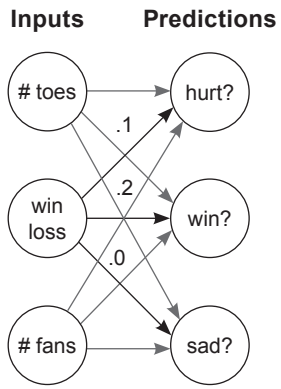



```
input = wlrec[0]
true = [hurt[0], win[0], sad[0]]
pred = neural_network(input,weights)

error = [0, 0, 0]
delta = [0, 0, 0]

for i in range(len(true)):
    error[i] = (pred[i] - true[i]) ** 2
    delta[i] = pred[i] - true[i]
weight_deltas = scalar_ele_mul(input,weights)
alpha = 0.1

for i in range(len(weights)):
    weights[i] -= (weight_deltas[i] * alpha)

print("Weights:" + str(weights))
print("Weight Deltas:" + str(weight_deltas))
```

Gradient descent with multiple inputs and outputs

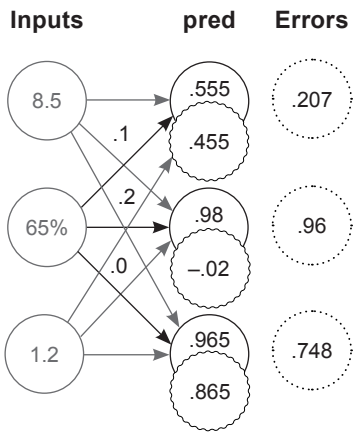
Gradient descent generalizes to arbitrarily large networks.

❶ An empty network with multiple inputs and outputs



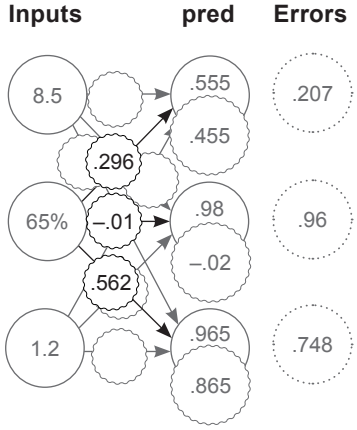

```
                 # toes %win # fans
weights = [ [0.1, 0.1, -0.3],# hurt?
            [0.1, 0.2, 0.0], # win?
            [0.0, 1.3, 0.1] ]# sad?

def vect_mat_mul(vect,matrix):
    assert(len(vect) == len(matrix))
    output = [0,0,0]
    for i in range(len(vect)):
       output[i] = w_sum(vect,matrix[i])
    return output

def neural_network(input, weights):

   pred = vect_mat_mul(input,weights)

   return pred
```

❷ PREDICT: Making a prediction and calculating error and delta




```
toes =  [8.5, 9.5, 9.9, 9.0]
wlrec = [0.65,0.8, 0.8, 0.9]
nfans = [1.2, 1.3, 0.5, 1.0]

hurt  = [0.1, 0.0, 0.0, 0.1]
win   = [  1,   1,   0,   1]
sad   = [0.1, 0.0, 0.1, 0.2]

alpha = 0.01

input = [toes[0],wlrec[0],nfans[0]]
true  = [hurt[0], win[0], sad[0]]


pred = neural_network(input,weights)

error = [0, 0, 0]
delta = [0, 0, 0]

for i in range(len(true)):

    error[i] = (pred[i] - true[i]) ** 2
    delta = pred[i] - true[i]
```

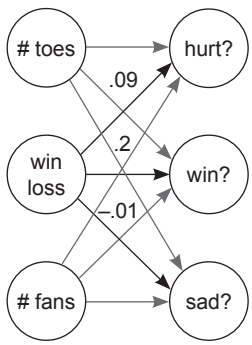
❸ COMPARE: Calculating each weight_delta and putting it on each weight



(`weight_delta`s are shown for only one input, to save space.)

```
def outer_prod(vec_a, vec_b):

    out = zeros_matrix(len(a),len(b))

    for i in range(len(a)):
        for j in range(len(b)):
            out[i][j] = vec_a[i]*vec_b[j]

    return out

input = [toes[0],wlrec[0],nfans[0]]
true = [hurt[0], win[0], sad[0]]

pred = neural_network(input,weights)

error = [0, 0, 0]
delta = [0, 0, 0]

for i in range(len(true)):

    error[i] = (pred[i] - true[i]) ** 2
    delta = pred[i] - true[i]

weight_deltas = outer_prod(input,delta)
```

❹ LEARN: Updating the weights




```
input = [toes[0],wlrec[0],nfans[0]]
true = [hurt[0], win[0], sad[0]]

pred = neural_network(input,weights)

error = [0, 0, 0]
delta = [0, 0, 0]

for i in range(len(true)):

    error[i] = (pred[i] - true[i]) ** 2
    delta = pred[i] - true[i]

weight_deltas = outer_prod(input,delta)

for i in range(len(weights)):
  for j in range(len(weights[0])):
     weights[i][j] -= alpha * \
                      weight_deltas[i][j]
```

What do these weights learn?

Each weight tries to reduce the error, but what do they learn in aggregate?

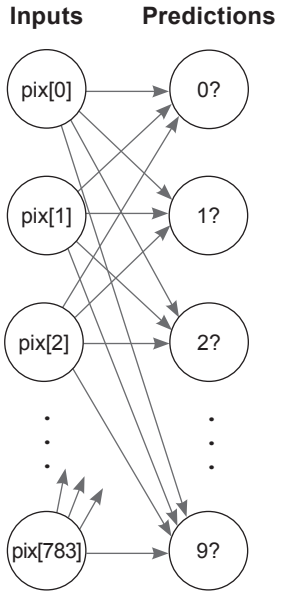
Congratulations! This is the part of the book where we move on to the first real-world dataset. As luck would have it, it's one with historical significance.

It's called the Modified National Institute of Standards and Technology (MNIST) dataset, and it consists of digits that high school students and employees of the US Census Bureau handwrote some years ago. The interesting bit is that these handwritten digits are black-and-white images of people's handwriting. Accompanying each digit image is the actual number they were writing (0–9). For the last few decades, people have been using this dataset to train neural networks to read human handwriting, and today, you're going to do the same.

Each image is only 784 pixels (28 × 28). Given that you have 784 pixels as input and 10 possible labels as output, you can imagine the shape of the neural network: each training example contains 784 values (one for each pixel), so the neural network must have 784 input values. Pretty simple, eh? You adjust the number of input nodes to reflect how many datapoints are in each training example. You want to predict *10 probabilities*: one for each digit. Given an input drawing, the neural network will produce these 10 probabilities, telling you which digit is most likely to be what was drawn.

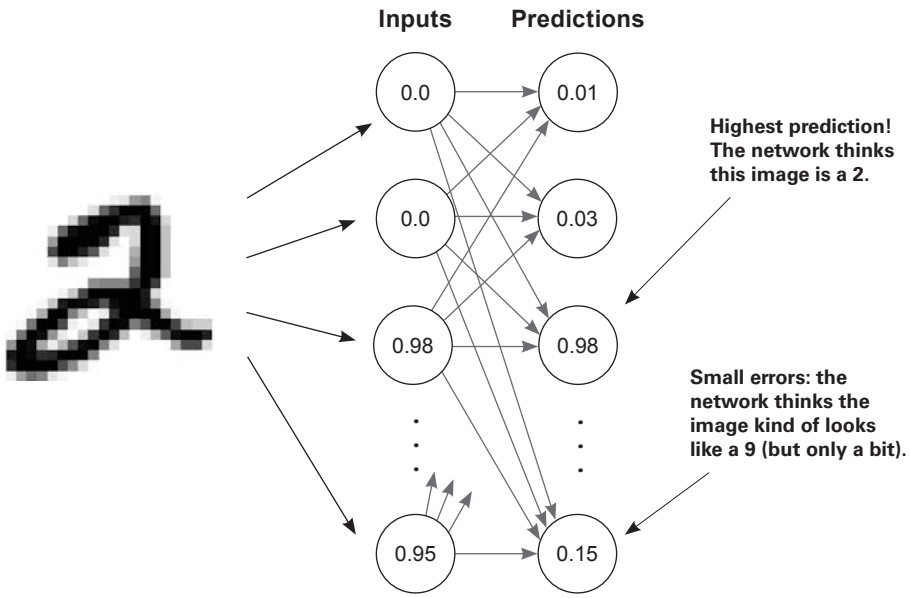
How do you configure the neural network to produce 10 probabilities? In the previous section, you saw a diagram for a neural network that could take multiple inputs at a time and make multiple predictions based on that input. You should be able to modify this network to have the correct number of inputs and outputs for the new MNIST task. You'll tweak it to have 784 inputs and 10 outputs.

In the MNISTPreprocessor notebook is a script to preprocess the MNIST dataset and load the first 1,000 images and labels into two NumPy matrices called `images` and `labels`. You may be wondering, "Images are two-dimensional. How do I load the (28 × 28) pixels into a flat neural network?" For now, the answer is simple: flatten the images into a vector of 1 × 784. You'll take the first row of pixels and concatenate them with the second row, and the third row, and so on, until you have one list of pixels per image (784 pixels long).



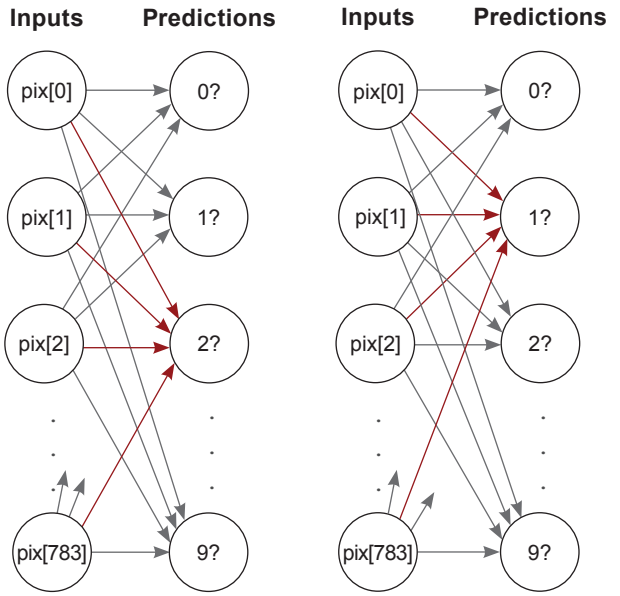
This diagram represents the new MNIST classification neural network. It most closely resembles the network you trained with multiple inputs and outputs earlier. The only difference is the number of inputs and outputs, which has increased substantially. This network has 784 inputs (one for each pixel in a 28 × 28 image) and 10 outputs (one for each possible digit in the image).

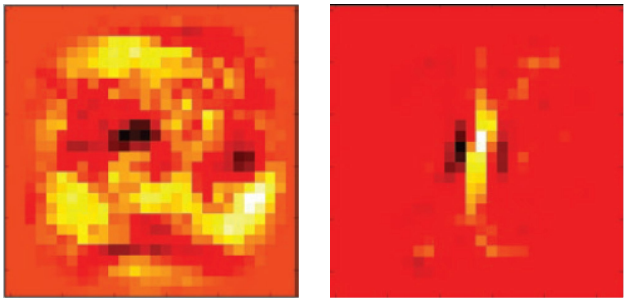
If this network could predict perfectly, it would take in an image's pixels (say, a 2, like the one in the next figure) and predict a 1.0 in the correct output position (the third one) and a 0 everywhere else. If it were able to do this correctly for all the images in the dataset, it would have no error.



Over the course of training, the network will adjust the weights between the input and prediction nodes so that `error` falls toward 0 in training. But what does this do? What does it mean to modify a bunch of weights to learn a pattern in aggregate?

Visualizing weight values





An interesting and intuitive practice in neural network research (particularly for image classifiers) is to visualize the weights as if they were an image. If you look at this diagram, you'll see why.

Each output node has a weight coming from every pixel. For example, the 2? node has 784 input weights, each mapping the relationship between a pixel and the number 2.

What is this relationship? Well, if the weight is high, it means the model believes there's a high degree of *correlation* between that pixel and the number 2. If the number is very low (negative), then the network believes there is a very low correlation (perhaps even negative correlation) between that pixel and the number 2.

If you take the weights and print them out into an image that's the same shape as the input dataset images, you can see which pixels have the highest correlation with a particular output node. In our example, a very vague 2 and 1 appear in the two images, which were created using the weights for 2 and 1, respectively. The bright areas are high weights, and the dark areas are negative weights. The neutral color (red, if you're reading this in the eBook) represents 0 in the weight matrix. This illustrates that the network generally knows the shape of a 2 and of a 1.

Why does it turn out this way? This takes us back to the lesson on dot products. Let's have a quick review.

Visualizing dot products (weighted sums)

Recall how dot products work. They take two vectors, multiply them together (elementwise), and then sum over the output. Consider this example:

```
a = [ 0, 1, 0, 1]
b = [ 1, 0, 1, 0]

    [ 0, 0, 0, 0] -> 0
```

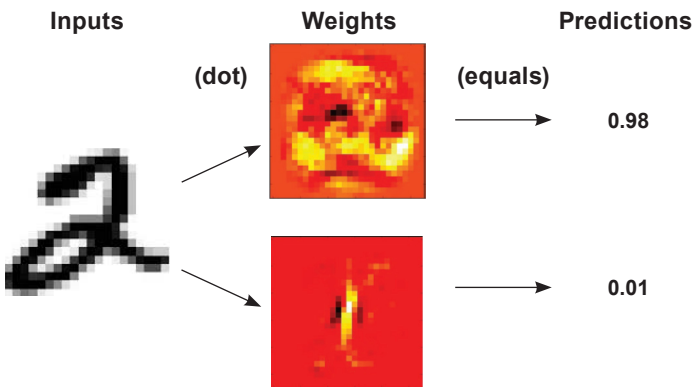
Score

First you multiply each element in `a` and `b` by each other, in this case creating a vector of 0s. The sum of this vector is also 0. Why? Because the vectors have nothing in common.

```
c = [ 0, 1, 1, 0]          b = [ 1, 0, 1, 0]
d = [.5, 0,.5, 0]          c = [ 0, 1, 1, 0]
```

But the dot products between `c` and `d` return higher scores, because there's overlap in the columns that have positive values. Performing dot products between two identical vectors tends to result in higher scores, as well. The takeaway? *A dot product is a loose measurement of similarity between two vectors.*

What does this mean for the weights and inputs? Well, if the `weight` vector is similar to the `input` vector for 2, then it will output a high score because the two vectors are similar. Inversely, if the `weight` vector is *not* similar to the `input` vector for 2, it will output a low score. You can see this in action in the following figure. Why is the top score (0.98) higher than the lower one (0.01)?



Summary

Gradient descent is a general learning algorithm.

Perhaps the most important subtext of this chapter is that gradient descent is a very flexible learning algorithm. If you combine weights in a way that allows you to calculate an error function and a `delta`, gradient descent can show you how to move the weights to reduce the error. We'll spend the rest of this book exploring different types of weight combinations and error functions for which gradient descent is useful. The next chapter is no exception.